D2 Design

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D2 Overview

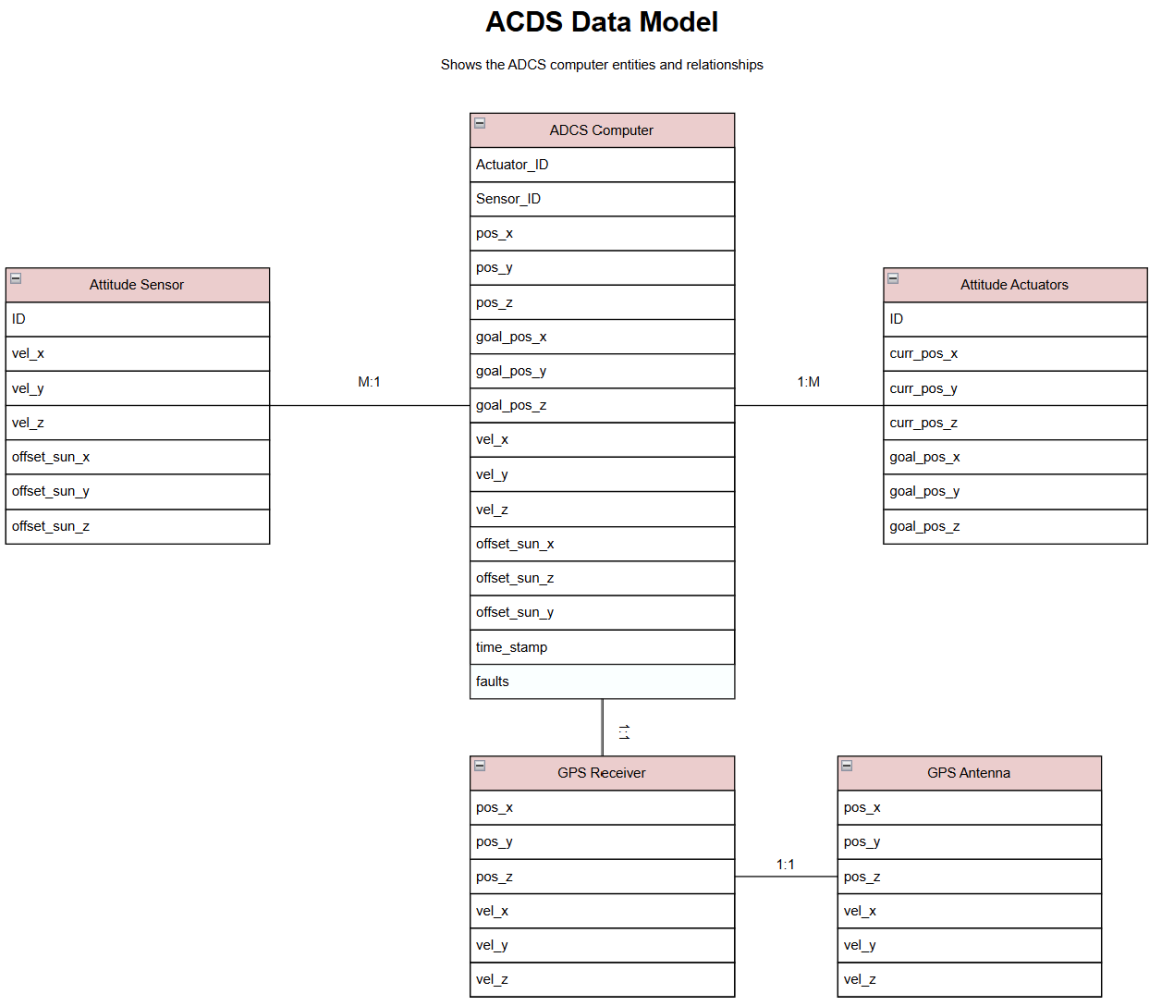
In collaboration with the CubeCats organization, this project focuses on the design, implementation, and verification of the initial prototype flight software for the On-Board Computer (OBC) on the HABSat-1 mission. The flight software shall establish core Command and Data Handling (CDH) capabilities across satellite subsystems and the ground station. It will also be able to execute autonomous and scheduled tasks across mission flight states.

D1 details the data structures and component interfaces for the Main Computer, ACDS Computer, Power Computer, Imager Adapter, Data Processor, and Data Storage.

D2 focuses on the Telemetry/Command Radio, Main Downlink Radio, Ground Station data pipeline, as well as the data models for the ACDS, Payload, Power, and Comms subsystems.

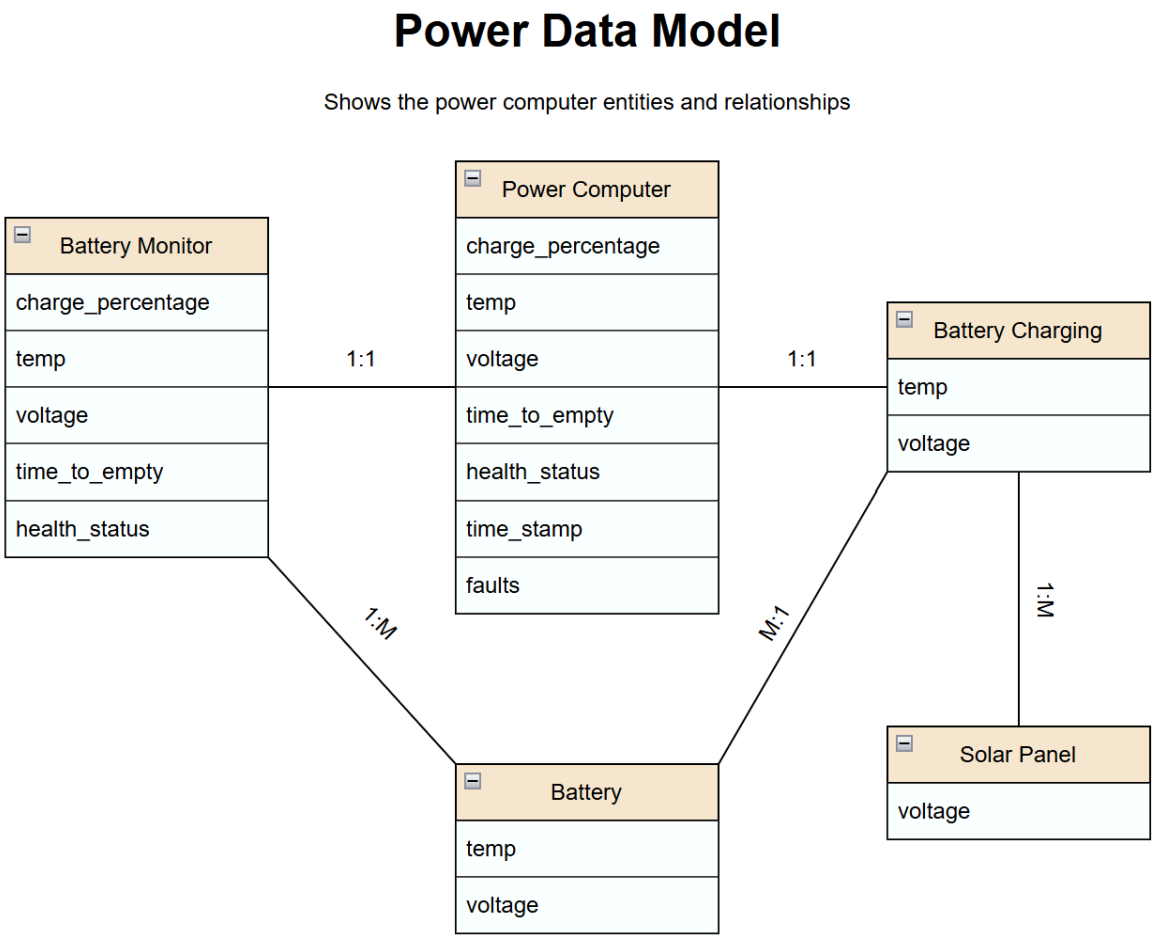
D2 diagrams

ACDS



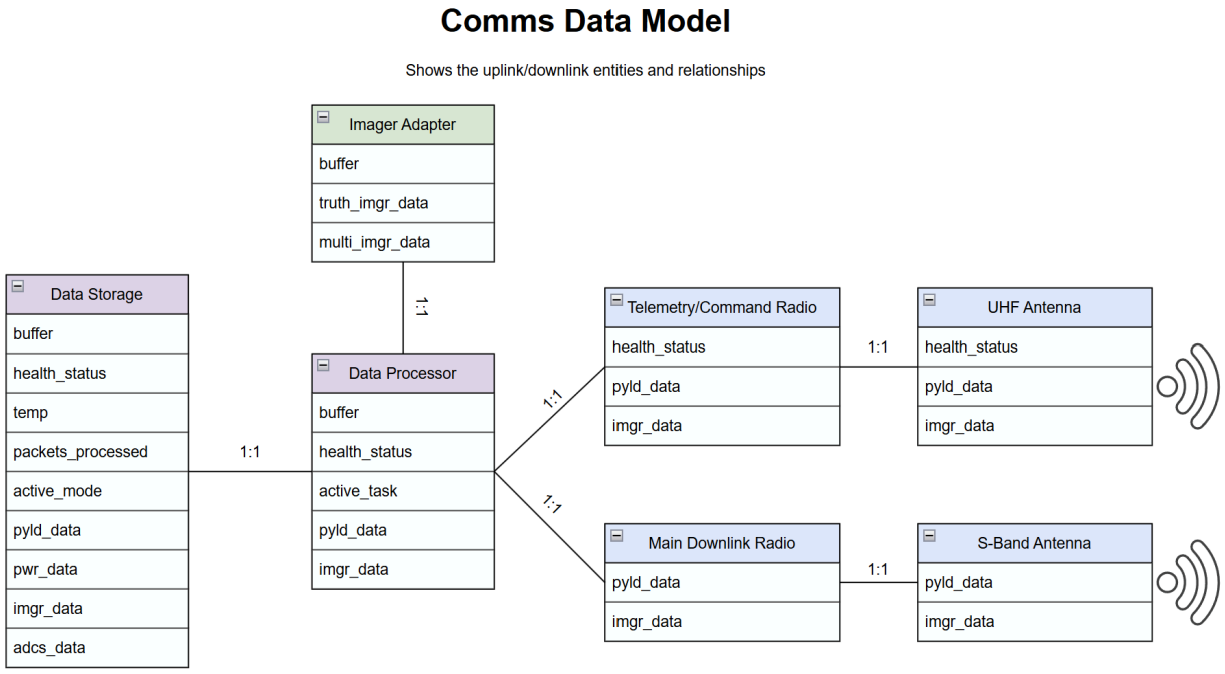
The relationships in this diagram that are a 1:1 cardinality, are due to the connections being between singular hardware components. The 1:M cardinalities are due to the connections being from one hardware component to multiple hardware components.

Power



The relationships in this diagram that are a 1:1 cardinality, are due to the connections being between singular hardware components. The 1:M cardinalities are due to the connections being from one hardware component to multiple hardware components.

Comms



All the relationships in this diagram are a 1:1 cardinality, as the connections are between singular hardware components.
The WiFi symbols represent the flow to an external source (ground station).

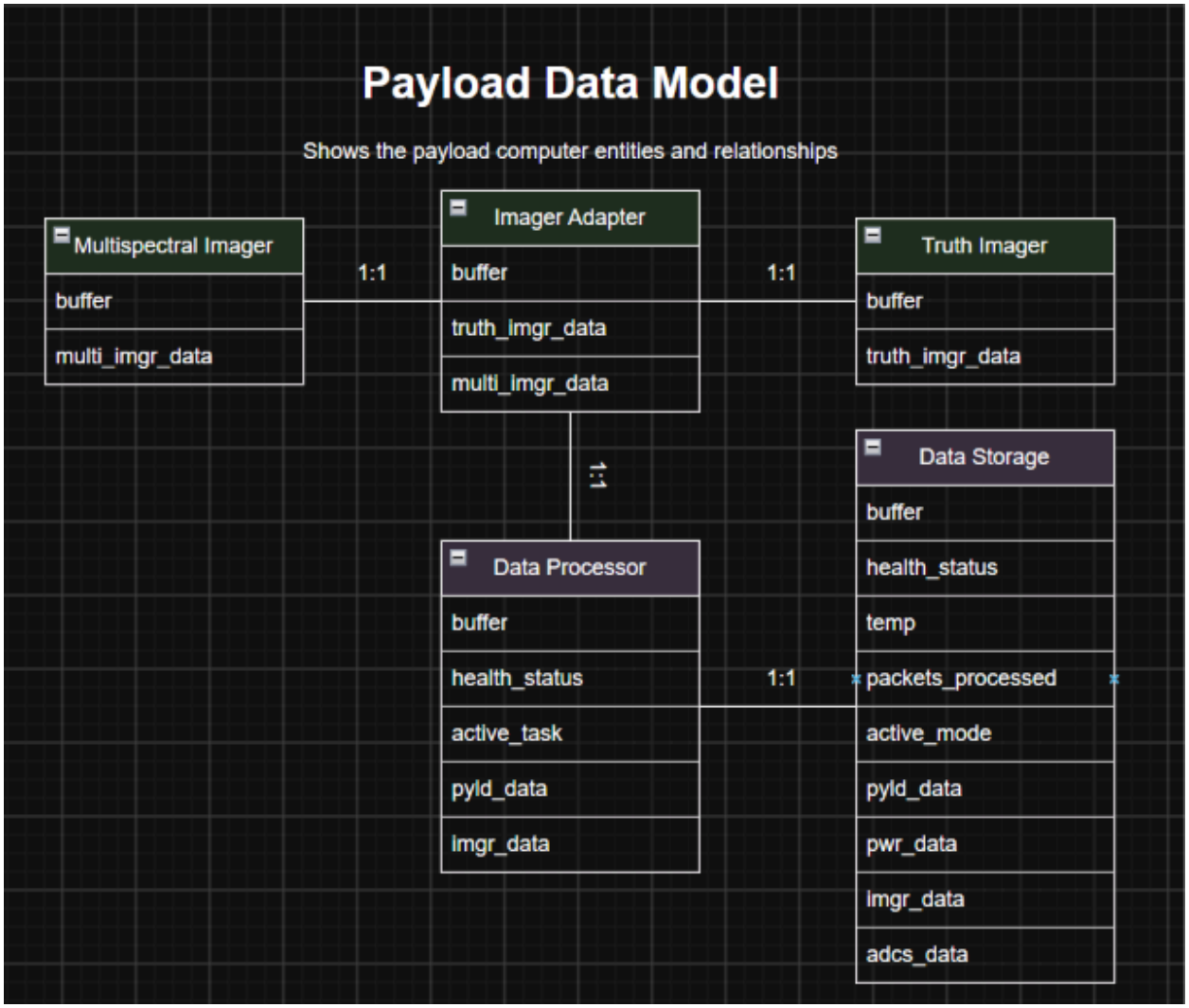
Command	ID	Encrypted	Used For:
Initiate Main Mission	0x00	TRUE	Uplinked during Heartbeat Mode to acknowledge initial ground contact, set MISSION_COMMISSIONED_FLAG = TRUE in non-volatile memory, and transition the satellite into Standby Mode to begin nominal orbital operations.
Transmit Telemetry	0x01	TRUE	Commands the satellite to downlink detailed real-time and recorded engineering telemetry (subsystem voltages, currents, temperatures, ADCS states, and flight software flags) over the primary UHF radio
Repeat Message	0x02	FALSE	Functions as an amateur radio digipeater relay, commanding the UHF radio to stage and retransmit received amateur radio frames across the authorized downlink band.
Exit Safe Mode	0x03	TRUE	Clears latched subsystem fault flags and commands the satellite to transition from Safe Mode back into Checkout Mode for automated diagnostics and recovery
Enter Low-Power Mode	0x04	TRUE	Manually overrides flight logic to force the satellite into Low-Power Mode, shedding non-essential payload power rails and commanding the ADCS into Sun-Pointing Mode to recharge

			batteries.
Power Cycle Systems	0x05	TRUE	Commands the Power Computer (EPS) to power cycle designated subsystem power rails (ADCS, Imagers, Data Processor) to clear bus lockups, single-event latch-ups, or uncommunicative states.
Reboot Flight Computer	0x06	TRUE	Triggers a controlled software reset and watchdog reboot of the Main Computer (OBC) to reload system state registers from persistent non-volatile storage
Enter RF Inhibit (Radio Silence)	0x07	TRUE	Enforces complete RF transmitter silence (disabling UHF and S-band transmission stages) for launcher deployment timer compliance, passivity windows, or thermal/power conservation
Exit RF Inhibit (Resume Radio)	0x08	TRUE	Restores power to the RF transmission stages, resuming nominal ground communication passes and the periodic 49-second housekeeping beacon broadcasts
Enter Emergency Mode	0x09	TRUE	Forces an immediate failsafe state across all subsystems, cutting high-draw buses, locking the ADCS into safe Sun-pointing orientation, and falling back to minimal carrier beaconing.
Exit Emergency Mode	0x0A	TRUE	Exits the emergency failsafe state after ground operations diagnose anomaly causes and verify battery and bus rail stability, returning the satellite to Standby or Checkout
Send Errors	0x0B	TRUE	Dumps recent error codes, watchdog reset counters, communication timeout records, and subsystem fault logs stored on the Data Storage module
Downlink Secondary Payload Packet	0x0F	TRUE	Queries and downlinks a specific, single telemetry or experiment data packet from the Secondary Payload module using its unique packet index.
Downlink Secondary Payload Range	0x10	TRUE	Queries and downlinks a sequential block of stored Secondary Payload data packets defined by start and end packet sequence numbers
Send Specified Images	0x11	TRUE	Commands the Data Processor to load processed chlorophyll index images or full-resolution multispectral/truth data from storage and stream them over the high-rate S-Band downlink radio
Uplink Orbit TLE Set	0x12	TRUE	Uplinks refreshed Two-Line Element (TLE) orbital ephemeris data to the Main Computer to be forwarded to the ADCS for on-board orbit propagation and pass schedule computation
Send Image via UHF	0x13	TRUE	Downlinks highly compressed image previews, thumbnails, or raw metadata across the primary UHF radio link as a low-bandwidth backup when S-band downlink is unavailable.
Purge Stored Image	0x14	TRUE	Deletes ground-verified images from flash memory buffers on the Data Storage module and Image Adapter to free capacity for subsequent imaging passes

Send Target Coordinates	0x15	TRUE	Uploads new target latitude, longitude, and elevation coordinates to the OBC and ADCS for scheduling future algal bloom imaging passes over designated lake sites.
Sync Time	0x16	TRUE	Backup in case the GPS component does not work to allow for time syncing of the Main Computer.

Command API from Ground Station to Satellite.

Payload



All the relationships in this diagram are a 1:1 cardinality, as the connections are between singular hardware components.

Edge Cases/ Error Handling

ACDS

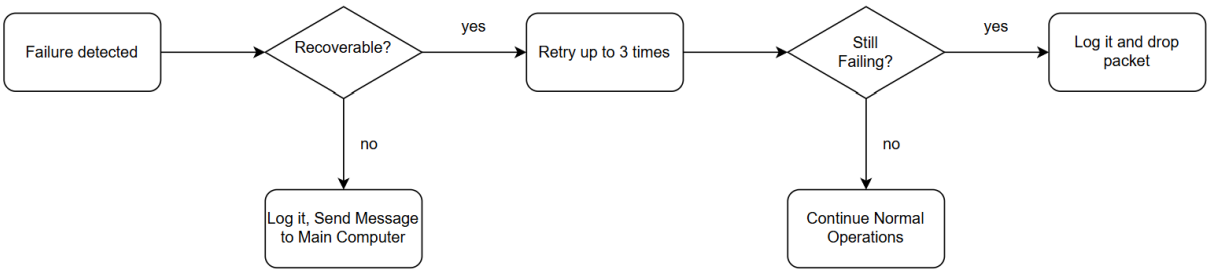
ID	Component	Category	Edge Case	Design Decision
1	ACDS Computer	Boundary values	Calculated orientation matrix value exceeds maximum floating-point limits (+Inf / NaN)	Fall back to the last valid state matrix. Handled by ACDS_MatrixGuard. Recoverable after telemetry state reset.
2	ACDS Computer	Timing and concurrency	Sensor polling thread starves attitude calculation loop cycle during high CPU load	Degrade precision by dropping non-critical sensor reads. Handled by ACDS_SchedulerManager. Recoverable next cycle.
3	ACDS Computer	Bad or missing input	Corrupted packet frame received over bus from attitude sensors	Retry reading up to 3 times before ignoring the frame. Handled by ACDS_BusParser. Recoverable on the next clean packet.
4	ACDS Computer	External dependencies	GPS receiver loses orbital lock during eclipse/shadowing	Fall back to dead reckoning using magnetometer and sun sensors. Handled by ACDS_NavFallback. Recoverable upon signal re-acquisition.
5	Attitude Sensor	Boundary values	Sun sensor reading saturates during direct solar flare alignment	Degrade filtering by masking saturated sensor output. Handled by SunSensor_Filter. Recoverable when flare subsides.
6	Attitude Sensor	Timing and concurrency	Gyroscope clock drift causes timestamp mismatch with accelerometer readings	Retry time sync over internal bus. Handled by Sensor_SyncHandler. Recoverable on pulse update.
7	Attitude Sensor	Bad or missing input	Coarse sun sensor returns zero	Surface an error flag to ACDS Computer and fall

			detected illumination due to body shadowing or solar eclipse	back to magnetometer and horizon sensors. Handled by SunSensor_Monitor. Recoverable after exiting shadow.
8	Attitude Sensor	External dependencies	Analog sensor reference voltage drops below 3.3V operating threshold	Surface an error status to power the bus and freeze sample output. Handled by Sensor_PowerSupervisor. Unrecoverable without hardware power reset.
9	Attitude Actuators	Boundary values	Reaction wheel requested RPM exceeds maximum hardware spin threshold	Degrade control torque by clamping max command value to threshold. Handled by Actuator_Limiter. Recoverable after momentum desaturation.
10	Attitude Actuators	Timing and concurrency	Magnetorquer actuation command overlaps with sensitive magnetometer sampling window	Fall back to time-division multiplexing to delay pulse. Handled by Actuator_TimingSync. Recoverable on the next frame slot.
11	Attitude Actuators	Bad or missing input	Null or NaN target torque vector received from flight computer	Surface an error and maintain the current momentum state. Handled by Actuator_CommandValidator. Recoverable upon receipt of valid torque command.
12	Attitude Actuators	External dependencies	Reaction wheel motor driver thermal sensor reports over-temperature fault	Degrade power to the wheel motor by 50%. Handled by Actuator_ThermalProtection. Recoverable after motor temperature drops.
13	GPS Receiver	Boundary values	Velocity calculation exceeds orbital threshold limits during fast maneuver	Degrade state confidence metric and clamp telemetry output. Handled by GPS_BoundsCheck. Recoverable when velocity stabilizes.

14	GPS Receiver	Timing and concurrency	1PPS (Pulse Per Second) interrupt signal missed by time manager	Fall back to the internal oscillator clock for timestamping. Handled by GPS_TimeKeeper. Recoverable on next valid 1PPS pulse.
15	GPS Receiver	Bad or missing input	NMEA sentences fail checksum verification	Retry reading buffer from serial interface. Handled by GPS_NMEAParser. Recoverable in the next sentence.
16	GPS Receiver	External dependencies	RF front-end active antenna bias circuit loses power	Surface an error flag to the main computer. Handled by GPS_HardwareMonitor. Unrecoverable without power rail toggle.
17	GPS Antenna	Boundary values	Signal-to-Noise ratio (SNR) falls below minimum decibel threshold	Degrade constellation tracking to strongest channels only. Handled by GPS_SignalQualityMonitor. Recoverable when SNR improves.
18	GPS Antenna	Timing and concurrency	Dual antenna switching logic triggers simultaneous toggle requests	Fall back to the default primary antenna channel. Handled by Antenna_SwitchArbiter. Recoverable on reset of toggle queue.
19	GPS Antenna	Bad or missing input	Antenna LNA reports zero current draw (open circuit/disconnected)	Surface an error to flight diagnostic log. Handled by Antenna_HealthCheck. Unrecoverable without physical connection restored.
20	GPS Antenna	External dependencies	Multi-path interference from satellite body reflection distorts phase	Degrade fix mode from 3D to 2D. Handled by GPS_MultipathFilter. Recoverable as geometry shifts.

ADCS Edge Case 3

Error handling decision flow for when an invalid sensor reading packet comes in.



Power

ID	Component	Category	Edge Case	Design Decision
21	Power Computer	Boundary values	Battery State of Charge (SoC) drops below critical 15% threshold	Fall back to Low-Power Safe Mode and shed non-critical payload loads. Handled by Power_SoCSupervisor. Recoverable when SoC reaches 40% via charging.
22	Power Computer	Timing and concurrency	Simultaneous power rail enable requests cause inrush current spike	Fall back to staggered switch sequence with 50ms delays. Handled by Power_Sequencer. Recoverable once rails stabilize.
23	Power Computer	Bad or missing input	Invalid telemetry packet read from power distribution bus	Retry read up to 3 times before taking the default safe power estimate. Handled by Power_Parser. Recoverable on clean read.
24	Power Computer	External dependencies	Solar array input voltage drops to zero during sudden eclipse transition	Fall back smoothly to battery power mode. Handled by Power_SourceSelector. Recoverable when exiting eclipse.
25	Battery Charging	Boundary values	Battery cell temperature exceeds safe	Degrade charge current to 0A (disable charging) while maintaining load lines.

			maximum charging limit (45°C)	Handled by Charge_ThermalCutoff. Recoverable when temp drops below 38°C.
26	Battery Charging	Timing and concurrency	MPPT algorithm update cycle conflicts with pulse-charging switch state	Retry MPPT tracking cycle after completing active charge pulse. Handled by MPPT_ControlLoop. Recoverable on the next tick.
27	Battery Charging	Bad or missing input	Corrupted voltage feedback value from charge controller ADC	Fall back to conservative trickle-charge rate and surface fault. Handled by Charge_ADCGuard. Recoverable when valid ADC data resumes.
28	Battery Charging	External dependencies	Solar panel regulator fails to provide minimum charge threshold voltage	Surface an error flag and isolate the charge circuit. Handled by Charge_InputMonitor. Recoverable when solar input voltage restores.
29	Battery Monitor	Boundary values	Current draw telemetry reading exceeds hardware shunt sensor maximum range	Surface an over-current alert immediately to the EPS protection switch. Handled by Monitor_OvercurrentGuard. Recoverable after fault clearing.
30	Battery Monitor	Timing and concurrency	Fuel gauge IC interrupt race condition during fast load transient switch	Retry reading register state after 10ms delay. Handled by Monitor_Battery_Arbiter. Recoverable on second attempt.
31	Battery Monitor	Bad or missing input	Coulomb counter reports impossible negative total capacity value	Fall back to the open-circuit voltage lookup table for SoC calculation. Handled by Monitor_CapacitySanity. Recoverable upon recalibration.
32	Battery Monitor	External dependencies	NTC thermistor sense line disconnects	Surface a sensor fault and assume conservative ambient temperature model. Handled by

			(reports -273°C reading)	Monitor_SensorFault. Unrecoverable without hardware fix.

Comms

ID	Component	Category	Edge Case	Design Decision
33	Telemetry/Command and Radio	Boundary values	Command buffer overflow due to oversized uplink command packet	Surface an error to ground station and discard packet. Handled by Radio_BufferManager. Recoverable on next command transmission.
34	Telemetry/Command and Radio	Timing and concurrency	Simultaneous uplink command reception and downlink telemetry transmission request	Fall back to half-duplex priority queuing (uplink preempts downlink). Handled by Radio_Arbiter. Recoverable after uplink processing completes.
35	Telemetry/Command and Radio	Bad or missing input	Failed decryption or HMAC verification on incoming command frame	Surface a security rejection alert and drop frame immediately. Handled by Radio_SecurityHandler. Recoverable on the next valid signed frame.
36	Telemetry/Command and Radio	External dependencies	Loss of ground station signal lock mid-burst transmission	Retry transmission up to 3 times before re-buffering payload. Handled by Radio_TxController. Recoverable on the next ground pass window.
37	Main Downlink Radio	Boundary values	Power amplifier temperature exceeds 85°C cutoff limit during high-rate downlink	Degrade data transmission rate by 50% to lower amplifier heat. Handled by Downlink_ThermalControl. Recoverable when the amplifier cools below 70°C.
38	Main Downlink Radio	Timing and concurrency	DMA buffer transfer underrun during payload streaming session	Fall back to idle frame insertion until buffer refills. Handled by Downlink_StreamManager.

				Recoverable automatically when DMA catches up.
39	Main Downlink Radio	Bad or missing input	Imager adapter supplies truncated payload file block	Retry requesting payload block from storage. Handled by Downlink_BlockRequester. Recoverable on successful fetch.
40	Main Downlink Radio	External dependencies	S-band transmitter synthesis lock fails due to clock reference instability	Surface an error flag and power-cycle RF synthesizer board. Handled by Downlink_LockMonitor. Recoverable post power-cycle.
41	UHF Antenna	Boundary values	VSWR exceeds 3:1 safety boundary due to antenna detuning	Degrade RF output power to prevent PA damage. Handled by UHF_VSWRGuard. Recoverable when detuning clears.
42	UHF Antenna	Timing and concurrency	Antenna burn-wire deployment pulse triggers during active transmission	Fall back to inhibiting RF output during the 5-second burn cycle. Handled by UHF_DeploySafety. Recoverable after the burn cycle ends.
43	UHF Antenna	Bad or missing input	Deployment sense switch reports indeterminate state (neither open nor closed)	Retry deployment burn sequence up to 2 times. Handled by UHF_DeployVerifier. Recoverable if the element deploys cleanly.
44	UHF Antenna	External dependencies	Atmospheric ionospheric disturbance causes total signal blackout	Surface loss-of-signal state and defer transmissions to queue. Handled by UHF_LinkMonitor. Recoverable as ionospheric conditions normalize.
45	S-Band Antenna	Boundary values	Off-axis steering angle exceeds 45° gain limit relative to ground station	Degrade modulation coding scheme to lower rate (BPSK). Handled by SBand_ModCodAdapter.

				Recoverable when tracking angle re-aligns.
46	S-Band Antenna	Timing and concurrency	Antenna pointing update calculation delayed by ADCS priority thread	Fall back to the last predicted orbital ephemeris vector. Handled by SBand_PointPredictor. Recoverable on the next ADCS update cycle.
47	S-Band Antenna	Bad or missing input	RF switch matrix control registers return unmapped/corrupted bitmask	Retry register, write and re-read verification. Handled by SBand_SwitchController. Recoverable after register reset.
48	S-Band Antenna	External dependencies	Structure thermal expansion causes patch feed impedance mismatch	Degrade RF output power level slightly to avoid reflective damage. Handled by SBand_ThermalComp. Recoverable when equilibrium returns.

Payload

ID	Component	Category	Edge Case	Design Decision
49	Imager Adapter	Boundary values	Image frame memory buffer allocation request exceeds maximum available RAM	Degrade resolution of active capture task. Handled by ImagerAdapter_MemAlloc. Recoverable when buffer memory is released.
50	Imager Adapter	Timing and concurrency	Simultaneous read request from Truth Imager and Multispectral Imager interfaces	Fall back to round-robin polling arbiters. Handled by ImagerAdapter_Arbiter. Recoverable continuously per frame cycle.
51	Imager Adapter	Bad or missing input	Missing line-sync pulse in raw image stream from sensor array	Retry re-synchronizing the frame line up to 3 times before dropping the corrupted frame. Handled by ImagerAdapter_SyncGuard. Recoverable on next frame start.

52	Imager Adapter	External dependencies	Data Processor bus interface stalls mid-transfer	Surface an error flag and flush the current local adapter buffer. Handled by ImagerAdapter_BusMonitor. Recoverable after bus interface reset.
53	Truth Imager	Boundary values	Pixel integration time requested exceeds sensor hardware register max (10,000ms)	Degrade setting by capping exposure to maximum hardware limit. Handled by TruthImager_ExposureControl. Recoverable on valid exposure command.
54	Truth Imager	Timing and concurrency	Trigger command arrives while previous exposure cycle is still actively reading out	Retry the queuing command until the current readout finishes. Handled by TruthImager_StateSequencer. Recoverable when readout completes.
55	Truth Imager	Bad or missing input	Hot pixel map configuration calibration file missing or corrupted on boot	Fall back to the default factory calibration table and surface warning. Handled by TruthImager_CalibLoader. Recoverable upon reloading valid map.
56	Truth Imager	External dependencies	Payload thermal control loop fails causing sensor plane to reach 60°C	Surface thermal critical alert and power down image sensor array. Handled by TruthImager_ThermalProtect. Recoverable after cooling.
57	Multispectral Imager	Boundary values	Spectral band filter wheel position sensor reads out of valid range bounds	Retry homing sequence for filter wheel motor. Handled by MultiImg_FilterHoming. Recoverable upon successful home indexing.
58	Multispectral Imager	Timing and concurrency	Multi-band frame capture trigger out of phase with satellite ground track timing	Degrade spatial overlap parameter and resynchronize capture clock. Handled by MultiImg_SyncControl.

				Recoverable on next track pass.
59	Multispectral Imager	Bad or missing input	Invalid gain value set in band command struct	Fall back to 1x default unity gain setting. Handled by MultiImg_ParamValidator. Recoverable when valid gain value is commanded.
60	Multispectral Imager	External dependencies	Regulated 12V payload power rail voltage drops below 10.8V limit	Surface an error flag and safely abort active spectral scan. Handled by MultiImg_PowerSupervisor. Recoverable when rail voltage recovers.
61	Secondary Payload	Boundary values	Payload sensor reading exceeds maximum allowable threshold	Clamp input value to maximum register limit. Handled by SecPayload_DataClipper. Recoverable when input returns to normal operating bounds.
62	Secondary Payload	Timing and concurrency	Payload data transfer window conflicts with high-priority system tasks	Buffer secondary payload data locally until bus bandwidth becomes available. Handled by SecPayload_CacheManager. Recoverable on the next idle bus cycle.
63	Secondary Payload	Bad or missing input	Invalid or corrupted command sequence received for secondary payload	Reject execution and log command error status. Handled by SecPayload_ScriptParser. Recoverable upon receipt of valid command frame.
64	Secondary Payload	External dependencies	Power rail supplying secondary payload is disabled due to system load shedding	Save active state configuration to non-volatile memory and enter standby. Handled by SecPayload_ShutdownHandler. Recoverable upon power restoration.

Main Computer

ID	Component	Category	Edge Case	Design Decision
65	Main Computer	Boundary values	RAM utilization exceeds critical threshold (95%) during heavy task processing	Degrade performance by terminating lowest priority background tasks. Handled by MainComp_MemManager. Recoverable as memory is freed.
66	Main Computer	Timing and concurrency	Watchdog timer tick starved due to continuous high-priority interrupt handling	Surface hard reset fault and trigger system reboot. Handled by MainComp_Watchdog. Recoverable post system reboot.
67	Main Computer	Bad or missing input	Non-volatile configuration memory fails CRC sanity check at boot	Fall back to golden image default parameters stored in ROM. Handled by MainComp_BootLoader. Recoverable after reflashing config partition.
68	Main Computer	External dependencies	CAN bus transceiver enters bus-off error state due to electrical noise	Retry resetting CAN controller interface logic up to 3 times. Handled by MainComp_CANSuperviso r. Recoverable when noise subsides.
69	Data Storage	Boundary values	Flash memory capacity reaches 99% full threshold	Fall back to overwriting oldest unflagged telemetry logs (FIFO). Handled by Storage_SpaceManager. Recoverable after data downlink and purge.
70	Data Storage	Timing and concurrency	Concurrent file write commands from Imager and Telemetry threads	Fall back to mutex locking and internal queue serialization. Handled by Storage_FileLockArbiter. Recoverable continuously per write task.
71	Data Storage	Bad or missing input	Bad flash block encountered	Retry writing to allocated spare block and mark original block bad. Handled

			during payload image chunk write	by Storage_BadBlockManage. Recoverable on successful re-map.
72	Data Storage	External dependencies	Storage drive controller drops off system SPI bus unexpectedly	Surface hardware error and fall back to onboard RAM log ring buffer. Handled by Storage_BusRecovery. Recoverable post interface re-initialization.
73	Data Processor	Boundary values	Payload image compression ratio calculation causes divide-by-zero	Fall back to raw uncompressed data transfer and log warning. Handled by Processor_MathGuard. Recoverable on next valid frame input.
74	Data Processor	Timing and concurrency	Image compression task execution time exceeds processing allocation slot	Degrade compression quality level to complete tasks faster. Handled by Processor_DynamicScheduler. Recoverable on next processing window.
75	Data Processor	Bad or missing input	Corrupted or incomplete frame data block received from Imager Adapter	Retry block request up to 3 times before discarding block. Handled by Processor_FrameVerifier. Recoverable on next frame block fetch.
76	Data Processor	External dependencies	Hardware DSP accelerator engine hangs during matrix processing	Surface error, reset DSP core, and fall back to CPU software processing. Handled by Processor_DSPMonitor. Recoverable after DSP reset.
77	Deployment Mechanisms	Boundary values	Current draw for deployment burn-wire exceeds max 2.5A driver threshold	Degrade driver current output via PWM duty cycle limiting. Handled by Deploy_CurrentLimiter. Recoverable when current normalizes.

78	Deployment Mechanisms	Timing and concurrency	Deployment pulse trigger command issued before solar panel switch arming delay expires	Fall back to queuing command until safety timer elapses. Handled by Deploy_SafetyTimer. Recoverable after timer expiry.
79	Deployment Mechanisms	Bad or missing input	Microswitch deployment status feedback signals contradictory states (both high and low)	Retry sampling microswitch state over 100ms debouncing window. Handled by Deploy_Debouncer. Recoverable upon stable state reading.
80	Deployment Mechanisms	External dependencies	Deployment driver enable line loses signal connection from GPIO board	Surface a deployment failure flag to ground telemetry. Handled by Deploy_GPIOWatchdog. Unrecoverable without redundant driver bus switch.

Requirements Traceability Table

ID	Requirement	Design element
1	Command Decryption	Main Computer, Telemetry/Command Radio, UHF Antenna, Radio_SecurityHandler (35)
2	Command Validation	Main Computer, Telemetry/Command Radio, UHF Antenna, Radio_SecurityHandler (35)
3	Image Downlink Operation	Main Computer, Main Downlink Radio, S-Band Antenna, Downlink_StreamManager (38), Downlink_BlockRequester (39)
4	Telemetry Downlink Operation	Main Computer, Telemetry/Command Radio, UHF Antenna, Radio_TxController (36)
5	Subsystem Control	Main Computer, RTOS scheduling
6	Subsystem Monitoring	Main Computer, Power_Parser (23), GPS_HardwareMonitor (16)

7	Heartbeat Operation	Main Computer, Telemetry/Command Radio, UHF Antenna
8	System Timekeeping	Main Computer, GPS_TimeKeeper (14), Deploy_SafetyTimer (78)
9	Storage Management	Main Computer, Data Storage, Storage_SpaceManager (69), Storage_FileLockArbiter (70), Storage_BadBlockManager (71)
10	Data Storage Telemetry	Data Storage, Storage_SpaceManager (69), Storage_BusRecovery (72)
11	Main Computer Telemetry	Main Computer, MainComp_MemManager (65), MainComp_Watchdog (66), MainComp_CANSupervisor (68)
12	Orientation Command	Main Computer, ACDS Computer, Attitude Actuators, SBand_PointPredictor (46)
13	Imaging Execution	Main Computer, Imager Adapter, Truth Imager, Multispectral Imager, TruthImager_StateSequencer (54)
14	Amateur Repeater Operation	Main Computer, Telemetry/Command Radio, UHF Antenna, Radio_Arbiter (34)
15	OBC Fault Recovery	Main Computer, MainComp_Watchdog (66), MainComp_BootLoader (67)
16	OBC Safeing	Main Computer, Power_SoCSupervisor (21)
17	OBC Unsafeing	Main Computer, Telemetry/Command Radio
18	Radiation-Resistant Reboot	Main Computer, MainComp_BootLoader (67)
19	EPS Solar Charging Efficiency	Main Computer, Power Computer, Solar Panel, Battery Charging, MPPT_ControlLoop (26)
20	Power Rail Regulation	Power Computer, Power Distribution Bus, Power_Sequencer (22)
21	Battery Thermal Cutoff Protection	Power Computer, Battery Charging, Charge_ThermalCutoff (25)
22	ADCS Pointing Accuracy	ACDS Computer, Attitude Sensor, Attitude Actuators, ACDS_MatrixGuard (1)
23	Payload Image Processing	Data Processor, Data Storage, Processor_MathGuard (73), Processor_DynamicScheduler (74)
24	Comms Thermal Throttling	Main Computer, Main Downlink Radio, Downlink_ThermalControl (37)

25	Secondary Payload Capture	Main Computer, Secondary Payload, SecPayload_CacheManager (62), SecPayload_ScriptParser (63), SecPayload_DataClipper (61), SecPayload_ShutdownHandler (64)

Design Rationale

Centralized Event-Driven Sequencer on Main Computer over Direct ADCS-to-Imager Triggering

We choose a centralized pass execution sequencer on the Main Computer over direct ADCS-to-Image Adapter triggering because direct triggering would bypass the power and health checks managed by the Main Computer. We accept the tradeoff of increased state machine complexity on the Main Computer because centralized safety enforcement prevents hardware damage during imaging passes.

Automatic Exclusion of Certain Passes based on Ground Station TLE validity checks

We choose to automatically exclude planned imaging passes when Ground Station TLE validity checks fail over leaving it completely up the the satellite because it will allow for less unnecessary processing on board. We accept this trade-off of additional complexity for the Ground Station because it will have less compute and time constraints and will allow for less unnecessary power being wasted on the satellite for passes that will not work.

Dual Radio Operations Redundancy over Single Radio Interface Architecture

We choose to configure both the Telemetry/Command Radio and Main Downlink Radio with cross-functional capability to perform critical communications over restricting each radio to dedicated single-purpose channels because establishing dual-radio operational redundancy prevents single-point RF failures from compromising mission command and telemetry link availability. We accept this tradeoff of managing additional state monitoring logic and dynamic interface switching because maintaining continuous command ability and telemetry downlink capability takes priority.

Dynamic UART Enable/Disable Control over Always-Active Serial Lines

We choose to implement dynamic UART interface enable/disable control on the communication channels between the Main Computer, Power Computer, and Telemetry/Command Radio based

on system operational state over maintaining always-active serial connections because conditional line toggling isolates internal component logic and prevents undesired command execution during restricted flight modes. We accept this tradeoff of increased firmware driver complexity because simplifying high-level subsystem interaction and state control logic improves overall system reliability.

Static Payload Command Queuing over Dynamic Memory Allocation

We choose to allocate fixed-size static memory buffers for payload command queuing over dynamic heap allocation because dynamic allocation introduces fragmentation and unpredictable execution jitter in embedded real-time systems. We accept the tradeoff of rigid command size limits because deterministic memory safety prevents out-of-memory kernel panics during critical orbital operations.

Verified Storage Handshake Buffer Purging over Immediate Post-Capture Memory Clearing

We choose to retain raw image files in onboard storage until explicitly commanded to purge by the Ground Station following confirmed reception because premature buffer clearance risks permanent loss of payload data during downlink failures. We accept the tradeoff of increased storage utilization because sufficient capacity is available to buffer multiple passes, ensuring mission-critical data can always be re-transmitted if a ground pass fails.

Peer Design Review Notes

Peer Review Summary

- **Criterion with Most Feedback: Completeness** and **Clarity** received the highest concentration of review comments, primarily concerning diagram component representations, missing persistent storage blocks, and explicit interface connections.
- **Top Two Issues Raised While Reviewing Others:**
    - Lack of explicit error recovery and fallback transitions in subsystem state diagrams.
    - Missing risk mitigation protocols for voltage drops across secondary payload power rails during continuous transmission passes.

ID	Rubric Criterion	Issue & Team	Action	Owner	Priority	Due Date
1	Clarity	Missing component coloring legend in	Add a color-coded legend defining	Kristen Griessman	Must-fix	10/21/26

		subsystem diagrams (Team DigiDarts & StreetSmart)	component classifications across all subsystem diagrams			
2	Feasibility	Direct connection between Battery and Battery Monitor missing in Power diagram (Team DigiDarts)	Add explicit 1:1 hardware/telemetry connection line between Battery and Battery Monitor	Elmoataz Abdellah	Must-fix	10/21/26
3	Completeness	MRAM non-volatile storage component missing from Data Storage diagram (Team StreetSmart)	Update Payload / Data Storage diagram to include MRAM block and bus interface	Seunghun Lee	Must-fix	10/21/26
4	Risk	Lack of watchdog supervisor coverage for dynamic UART serial interfaces (Team StreetSmart & SkillForge)	Define UART timeout watchdog and recovery handler logic in edge case table	Matthew Goldsberry	Must-fix	10/28/26